

Notes: Cao, Jiang, Wang & Yang (JFE, 2024)

From Man vs. Machine to Man + Machine: The art and AI of stock analyses

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1 Big picture

- **Question.** Can an AI analyst outperform human stock analysts, when do human analysts still win, and can analysts and AI together outperform either side alone?
- **Setting.** The paper builds an AI analyst that predicts 12-month stock returns using timely public information and compares it with I/B/E/S analyst target-price forecasts made on the same stock at the same time.
- **Main Man vs. Machine result.** The AI analyst beats human analysts for a majority of forecasts. Human analysts beat AI only 45.5% of the time, meaning AI wins 54.5% of the time.
- **Bias decomposition.** A machine-debiased version of human forecasts beats AI 46.5% of the time. Therefore, correctable analyst bias explains part of the human shortfall, but most of the Man–Machine gap comes from AI’s stronger information-processing ability.
- **Where humans win.** Humans retain an advantage for firms where institutional knowledge is important: smaller and less liquid firms, firms with more intangible assets, firms closer to distress, and firms in recessions or dynamic competitive environments.
- **Where AI wins.** AI performs better when information is transparent but voluminous. The number of information events is negatively related to human relative performance.
- **Man + Machine result.** A hybrid model that adds analyst forecasts to the AI information set outperforms both stand-alone analysts and stand-alone AI. The hybrid beats AI in 57.8% of the cases and beats stand-alone analysts in 54.8% of the cases.
- **Extreme error result.** Combining analysts with AI greatly reduces large forecast mistakes. At the 90th-percentile error threshold, Man + Machine avoids 90.7% of analyst-only extreme errors and 43.6% of AI-only extreme errors, while creating extreme errors in only 0.1% of previously safe cases.
- **Event-study result.** After alternative data become available, analysts whose brokerage houses have AI capability close part of the gap with the AI model.
- **Economic takeaway.** The paper does not argue that AI simply replaces skilled analysts. It shows that humans and machines have different comparative advantages, and that the best equilibrium is often augmentation rather than substitution.

2 Data and measurement

2.1 Forecast target

- The main prediction target is the 12-month stock return inferred from analysts’ 12-month target prices.
- The authors focus on returns rather than earnings because earnings can be managed or adjusted by firms, while target prices are closer to the investment decision that analysts help investors make.
- For each analyst forecast on stock i at date t , the AI model generates a forecast for the same stock and timing.
- The realized outcome is the ex post 12-month stock return.

Interpretation: This design makes the comparison direct: the human and the AI are forecasting the same object, for the same firm, at the same time.

2.2 Human analyst data

- Human forecasts come from I/B/E/S analyst target-price forecasts over 2001–2018.

- The full Man vs. Machine comparison covers 922,157 analyst forecasts.
- The main regression sample with complete firm-level, industry-level, macroeconomic, and alternative-data variables contains 352,358 observations.
- Analyst-level variables include analyst experience, brokerage affiliation, number of analysts in the brokerage firm, and star-analyst status.

Interpretation: The paper treats each analyst forecast as an observation, but also controls for firm, analyst, and year fixed effects in the relative-performance regressions.

2.3 AI information set

- The AI model uses information available before the forecast date.
- Predictors are grouped into six broad classes:
 - firm returns and prices;
 - firm characteristics;
 - earnings variables;
 - industry variables;
 - macroeconomic variables;
 - textual variables from disclosures, news, and social media.
- The AI deliberately excludes current and past analyst forecasts, so that it does not learn directly from analyst opinions.

Interpretation: Excluding analyst forecasts makes the Man vs. Machine comparison conservative: the AI cannot simply imitate analysts.

2.4 Machine-learning models

- The paper considers several standard machine-learning methods, including penalized regression, support vector machines, random forests, gradient boosting, and long short-term memory neural networks.
- The main AI analyst is an ensemble model that combines complementary prediction methods.
- The final AI forecast is designed to be out-of-sample and time-adapted: models are trained on past data and used to forecast future observations.

Interpretation: The AI is not a single black-box algorithm selected for a convenient result. It is an ensemble benchmark meant to approximate current professional AI practice in asset management.

2.5 Key outcome measures

For each human forecast and its concurrent AI forecast, the paper defines:

- **Analyst Beats AI:** equals one if the analyst has a smaller absolute prediction error than the AI;
- **Forecast Error Difference:** AI error minus analyst error, scaled by the maximum of the two errors; positive values favor the analyst;
- **Disagreement:** the normalized squared difference between the analyst forecast and the AI forecast;
- **Human Wins:** equals one when the analyst beats AI conditional on high disagreement;

- **Analyst + AI Beats AI:** equals one when the hybrid forecast has a smaller error than AI alone.

Interpretation: These measures separate three questions: who is more accurate, when do humans and AI disagree, and when does the combination add incremental value.

2.6 Alternative-data event study

- The paper studies the staggered introduction of alternative data, especially satellite-imaging data that can produce real-time consumer-footprint information.
- *Alt Data Covered* equals one when the firm is covered by such data.
- *Post* equals one after the data become available.
- *AI Hiring* measures whether the analyst’s brokerage has AI capability, based on AI-related job postings.

Interpretation: Alternative-data adoption is used as a quasi-event to ask whether human analysts can catch up when they receive data that are useful only if their organization has enough AI capability to process it.

3 Models and empirical specifications

3.1 AI return prediction

For stock i at date t , the realized 12-month return is modeled as

$$R_{i,t,T} = R_{i,t}^{AI} + \varepsilon_{i,t}, \quad R_{i,t}^{AI} = f_{t-}(X_{i,t-}), \quad (1)$$

where $X_{i,t-}$ is the public information set available before date t , and $f_{t-}(\cdot)$ is trained only on past data.

Interpretation: The AI forecast is a direct prediction of future returns using only pre-forecast information. The notation emphasizes out-of-sample prediction.

3.2 Man vs. Machine accuracy

Let $R_{i,j,t}^{Man}$ be analyst j ’s forecast-implied return for stock i at date t . The analyst beats the AI when

$$|R_{i,j,t}^{Man} - R_{i,t,T}| < |R_{i,t}^{AI} - R_{i,t,T}|. \quad (2)$$

Interpretation: The comparison is pairwise. Every human forecast is matched against the AI forecast for the same firm and date.

3.3 Machine-debiased Man

The paper estimates predictable analyst forecast errors as

$$\log(1 + R_{i,j,t}^{Man}) - \log(1 + R_{i,t,T}) = g(X_{i,t-}, Z_{i,j,t-}) + e_{i,j,t}, \quad (3)$$

where $Z_{i,j,t-}$ includes analyst and brokerage characteristics. The machine-debiased forecast subtracts the predicted bias from the original analyst forecast.

Interpretation: If the debiased analyst still underperforms AI, then AI’s advantage is not just that it avoids correctable human biases. The paper finds exactly this.

3.4 Relative-performance regression

The determinants of human relative performance are estimated as

$$RelativePerformance_{i,j,t} = X'_{i,j,t}\beta + \alpha_i + \alpha_j + \alpha_{year} + \varepsilon_{i,j,t}, \quad (4)$$

where the dependent variable is either *Analyst Beats AI* or *Forecast Error Difference*.

Interpretation: The coefficients show the conditions under which humans beat AI after controlling for firm, analyst, and time effects.

3.5 Man + Machine hybrid

The hybrid model adds analyst forecasts to the machine information set:

$$R_{i,j,t}^{M+M} = h_{t-}(X_{i,t-}, F_{i,j,t-}), \quad (5)$$

where $F_{i,j,t-}$ contains analyst forecasts and recent analyst-performance information.

Interpretation: If analyst forecasts contain institutional knowledge not already captured by public data, the hybrid forecast should beat AI alone.

3.6 Alternative-data event study

The event-study specification is

$$AnalystBeatsAI_{i,j,t} = \beta_1(Treat_{i,t} \times AIHiring_{j,t}) + \beta_2 AIHiring_{j,t} \quad (6)$$

$$+ \beta_3 AltDataCovered_i + \beta_4 Post_t + Controls_{i,j,t} + \alpha_i + \alpha_{year} + \varepsilon_{i,j,t}, \quad (7)$$

where $Treat_{i,t} = AltDataCovered_i \times Post_t$.

Interpretation: The key coefficient is β_1 . A positive value means analysts benefit from alternative data only when their brokerage has AI capability.

4 Identification and empirical design

4.1 Why the main comparison is informative

- The AI and the human predict the same stock return at the same time.
- The AI uses only information available before the forecast and excludes analyst forecasts.
- The performance measure is based on realized returns, not on ex ante ratings or subjective forecast quality.
- The paper also evaluates return portfolios formed from AI-versus-human disagreement.

Interpretation: The main Man vs. Machine test is a clean benchmarking exercise, not a causal design. Its strength is the matched prediction environment.

4.2 Separating bias from information processing

- Analysts can underperform because of incentive or behavioral biases.
- They can also underperform because they cannot process large volumes of data as well as AI.
- The machine-debiased analyst forecast removes predictable forecast errors using analyst and firm information.
- Debiasing raises human performance from 45.5% to 46.5%, but humans still trail AI.

Interpretation: Only a modest part of the AI advantage comes from removing human bias. The larger part comes from information-processing capacity.

4.3 Comparative advantage tests

- The paper predicts humans will do better when institutional knowledge matters.
- It predicts AI will do better when the information environment is transparent but data-intensive.
- Firm characteristics such as illiquidity, intangible assets, size, distance to default, number of information events, and industry recession are used to test these predictions.

Interpretation: These tests do not only ask who wins; they explain what kind of task belongs to humans and what kind belongs to machines.

4.4 Why the Man + Machine result matters

- If AI fully subsumed analyst information, adding analyst forecasts would not improve prediction.
- The hybrid model improves accuracy, which means analysts contain incremental information.
- The hybrid also reduces extreme mistakes, which is important in high-stakes forecasting settings.

Interpretation: The paper's central message is complementarity. AI has superior average performance, but humans still add information that can make AI better.

4.5 Limitations

- The AI analyst is a benchmark created by the researchers, not necessarily the exact model used by any market participant.
- The main Man vs. Machine comparison is not randomized; it is an out-of-sample prediction comparison.
- The alternative-data event study is more causal in spirit, but AI hiring and data adoption are still organizational choices.
- The results are for stock analysis and 12-month return forecasts; they need not generalize mechanically to every expert profession.

Interpretation: The evidence is strongest for mapping relative predictive advantages. It is less suited to estimating a universal causal effect of AI on jobs.

5 Tables and figures

The visuals below are directly cropped from the uploaded paper and grouped compactly to improve reading speed.

5.1 Table 1, Figure 1, and Figure 2: data and basic AI performance

Read:

- Table 1 reports summary statistics for the regression sample and alternative-data variables.
- Figure 1 shows human analysts beat AI 45.5% of the time over the full sample, so AI wins 54.5%.
- The human beat ratio is volatile across years and generally below the 50% line.

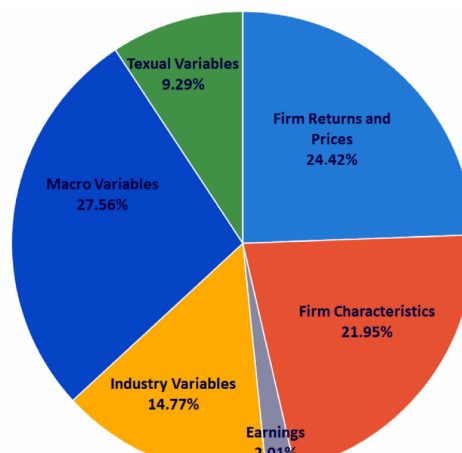
- Figure 2 decomposes the AI forecast contribution: macro variables contribute 27.6%, firm returns and prices 24.4%, firm characteristics 22.0%, industry variables 14.8%, textual variables 9.3%, and earnings only 2.0%.

Economic message: AI's edge comes from processing many public signals, especially macro, return, firm-characteristic, and industry information. Earnings alone are not the main source of AI performance.

Table 1
Summary statistics.
This table reports the summary statistics of key variables. The firm-level, industry-level, and macroeconomic variables are defined in Section 2.5. The AI and alternative data variables *AI Hiring*, *Alt Data Covered*, and *Post* are defined as follows. *textAI Hiring* is the ratio of the number of AI jobs to the total number of job postings. *Alt Data Covered* is an indicator variable equal to one if alternative data are available for the firm by the end of the sample, and if the firm is in an industry with retail footprints. *Post* is an indicator variable equal to one if a "treated" firm has been covered by alternative data. For "untreated" firms, *Post* is coded one if the year is after 2014. The mean, median, standard deviation, 25 percentile, 75 percentile, and number of observations for the panel data are reported in the table.

Variables	Mean	Median	Std	P25	P75	N
Panel A. Firm-level, industry-level, and macroeconomic variables						
<i>Amihud Illiquidity</i>	0.40	0.00	56.10	0.00	0.02	352 358
<i>Intangible Assets</i>	0.01	-0.13	1.11	-0.34	-0.04	352 358
<i># Analysts in Brokerage Firm</i>	0.03	0.04	0.01	0.03	0.04	352 358
<i>Star Analysts</i>	0.03	0.00	0.16	0.00	0.00	352 358
<i>Number of Information Events</i>	2.64	2.64	0.69	2.30	3.00	352 358
<i>Distance to Default</i>	5.44	5.20	2.64	3.52	6.80	352 358
<i>Market Cap</i>	8.07	7.98	1.68	6.87	9.21	352 358
<i>% Institutional Holdings</i>	0.66	0.77	1.85	0.53	0.90	352 358
<i>Fluidity</i>	7.19	6.48	3.66	4.50	9.14	352 358
<i>Text Complexity</i>	16.04	15.96	1.22	15.40	16.55	352 358
<i>Industry Recession</i>	0.11	0.00	0.31	0.00	0.00	352 358
Panel B. AI and alternative data variables						
<i>AI Job</i>	0.01	0.00	0.07	0.00	0.00	56 697
<i>Alt data cover</i>	0.02	0.00	0.40	0.00	0.00	56 390
<i>Post</i>	0.66	1.00	0.47	0.00	1.00	56 697

(a) Table 1: summary statistics



(b) Figure 2: variable-group contributions

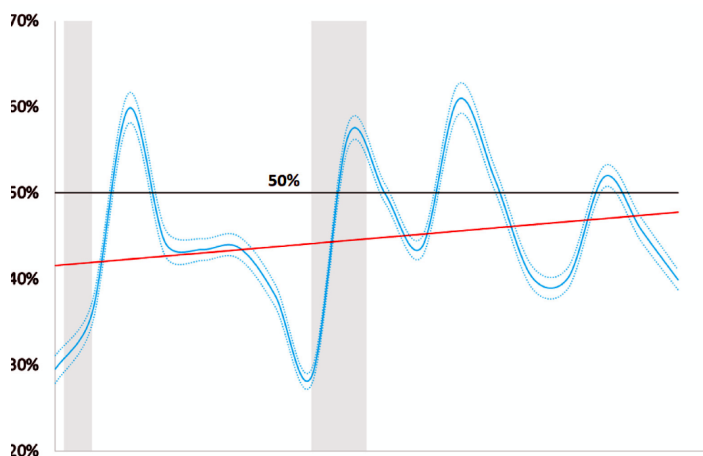


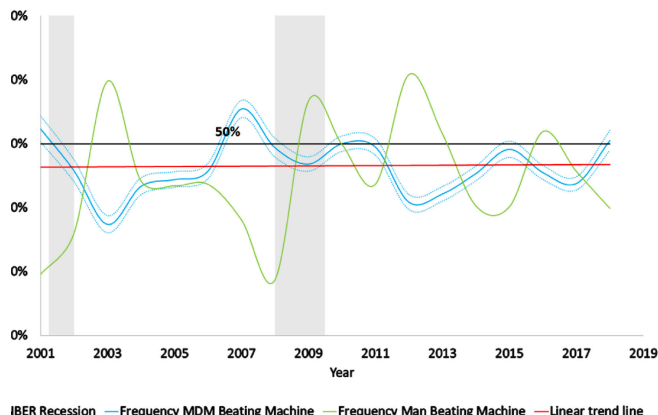
Figure 2: Figure 1: Man vs. Machine beat ratio

5.2 Figure 3, Table 2, and Table 3: debiasing, persistence, and portfolios

Read:

- Figure 3 shows machine-debiased human forecasts perform better than raw analyst forecasts, but they still usually trail AI.
- MDM beats AI 46.5% of the time, so debiasing explains about 22% of the human-machine gap.
- Table 2 shows past winners remain somewhat better than past losers, but even persistent top analysts barely reach the 50% beat benchmark.
- Table 3 shows portfolios based on AI-versus-human recommendations earn positive long-short returns and alphas, suggesting AI signals contain investment value beyond analyst signals.

Economic message: Correcting analyst bias improves humans slightly, but not enough. AI's advantage is not just behavioral bias; it is also data-processing power.



(a) Figure 3: MDM vs. AI

rences to color in this figure legend, the reader is referred to the web version of this article.)

Table 2
Persistence of performance of AI analyst.
Each year, analysts are sorted by mean squared prediction errors of log prices based on the past one, past two, and up to five years. If the mean squared error over the last year is below (above) the median during the specified past period, the analyst is in the top (bottom) in the current year. In Panel A, the sorting is based on the full period of the past one, two, ..., five years. In Panel B, the sorting requires that an analyst be in the top half in each of the past one, two, ..., five years to be placed in the "top" group. Both panels report the analyst beat ratio, i.e., the number of times analysts beat AI, as a proportion of total number of predictions.

Panel A: Analyst beat ratio sorted by analysts who are above/below median					
	1 year	2 years	3 years	4 years	5 years
Analyst top	49.25%	49.26%	49.11%	49.10%	49.08%
Analyst bottom	43.01%	42.95%	43.04%	43.04%	43.05%

Panel B: Analyst beat ratio sorted by analysts who are above median each of the past years					
	1 years	2 years	3 years	4 years	5 years
Analyst Persistent top	49.25%	50.33%	49.87%	49.63%	49.40%
Analyst Persistent bottom	43.01%	42.34%	41.39%	40.95%	40.79%

(b) Table 2: persistent analyst performance

Table 3
Portfolio performance following machine vs. man recommendations.
At the beginning of each month, we gather all predictions made by all analysts and the corresponding AI forecasts in the past 30, 60, 90, 180, and 360 days. For each pair of predictions, if the Machine's prediction is greater than the median of Machine predictions in the month and the human's prediction is less than the median Machine prediction in the month, we define it as a buy signal. When both conditions are negated, we define it as a sell signal. During the given time horizon, the portfolio will long the stock if there are more buy than sell signals and short the stock otherwise. The portfolios are value-weighted and rebalanced every six months, i.e., a position is held for six months or till the signals reverse. The monthly percentage returns of the long-short, long-leg (stocks only with a buy sign) and short-leg portfolios (stocks only with a short sign), as well as the alphas generated from the FF3, FFC4, FF5, and FF6 models, are presented. The OLS standard error is used to construct *t*-stats. The *t*-stats are reported in parentheses. ***, **, * denote statistical significance at the 0.01, 0.05, and 0.10 levels (two-tailed), respectively.

Portfolio returns — AI vs. Analyst						
Machine vs Human						
Long-Short		30 day inform	60 day inform	90 day inform	180 day inform	360 day inform
Monthly returns	Ret	0.64*** (3.06)	0.65*** (3.67)	0.58*** (3.33)	0.60*** (3.79)	0.53*** (3.66)
	FF3	0.69*** (3.40)	0.72*** (4.19)	0.64*** (3.80)	0.65*** (4.24)	0.61*** (4.45)
	FFC4	0.62*** (3.16)	0.65*** (3.97)	0.58*** (3.57)	0.58*** (4.03)	0.52*** (4.21)
	FF5	0.51** (2.54)	0.62*** (3.73)	0.55*** (3.37)	0.55*** (3.72)	0.51*** (3.74)
	FF6	0.51*** (2.62)	0.63*** (3.92)	0.56*** (3.54)	0.56*** (4.02)	0.50*** (4.09)
Long-Leg		30 day inform	60 day inform	90 day inform	180 day inform	360 day inform
Monthly returns	Ret	0.92*** (2.69)	0.97*** (2.96)	0.93*** (2.83)	0.97*** (3.07)	0.98*** (3.32)
	FF3	0.42*** (3.12)	0.48*** (4.12)	0.44*** (3.99)	0.44*** (4.53)	0.42*** (5.09)
	FFC4	0.39*** (2.93)	0.47*** (3.98)	0.42*** (3.84)	0.41*** (4.34)	0.38*** (4.84)
	FF5	0.37*** (2.76)	0.46*** (3.90)	0.42*** (3.91)	0.40*** (4.28)	0.37*** (4.43)
	FF6	0.37*** (2.81)	0.46*** (3.92)	0.42*** (3.94)	0.40*** (4.40)	0.36*** (4.59)
Short-Leg		30 day inform	60 day inform	90 day inform	180 day inform	360 day inform
Monthly returns	Ret	-0.28 (-0.78)	-0.32 (-0.89)	-0.35 (-0.99)	-0.37 (-1.09)	-0.44 (-1.33)
	FF3	0.27** (2.49)	0.23*** (2.82)	0.20 (2.44)	0.21*** (2.83)	0.19*** (2.63)
	FFC4	0.23** (2.22)	0.18** (2.51)	0.16** (2.10)	0.17** (2.52)	0.13** (2.18)
	FF5	0.14 (1.27)	0.16*** (2.02)	0.13 (1.58)	0.15** (2.01)	0.14** (1.98)
	FF6	0.14 (1.30)	0.16** (2.29)	0.13* (1.74)	0.15** (2.27)	0.14** (2.17)

ials' performance. Further, Figure IA.1 shows a roughlv (mean square errors) of the forecasts by all analysts in the

Figure 4: Table 3: portfolio performance following machine vs. man recommendations

5.3 Figure 4, Table 4, and Figure 5: human comparative advantage and disagreement

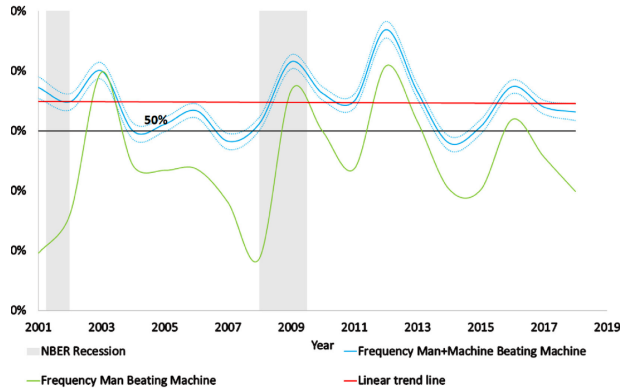
Read:

- Figure 4 shows the hybrid Man + Machine analyst beats stand-alone AI in most years.
- Table 4 shows humans perform better for illiquid firms, firms with more intangible assets, firms covered by

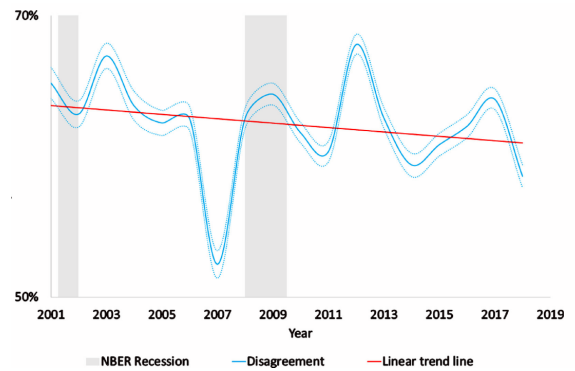
larger brokerages, firms closer to distress, and firms in industry recessions.

- Table 4 also shows AI performs better when the number of information events is high and when firms are larger or more transparent.
- Figure 5 shows human–AI disagreement tends to decline over time, consistent with analysts absorbing more big-data and AI-related tools.

Economic message: Humans are not obsolete. They are most useful where context, soft information, and institutional knowledge matter. AI dominates when the task is to digest large volumes of transparent information.



(a) Figure 4: AI-assisted analysts vs. AI



(b) Figure 5: Man–Machine disagreement

Table 4

Man vs. Machine: The relative advantage of analyst vs AI.

This table presents the coefficients and *t*-stats of regressing the *Analyst Beats AI* indicator (Panel A) and the *Forecast Error Difference: Analyst vs. AI* (Panel B) on the firm-level, industry-level, and macroeconomic variables presented in Table 1. *Analyst Beats AI* is an indicator variable equal to one if the analyst beats the AI. *Forecast Error Difference: Analyst vs. AI* is defined as the difference between absolute prediction errors between the AI and the analysts, divided by the maximum value of these two prediction errors. The number is positive if the analyst has a smaller absolute forecast error, i.e., the analyst beats AI. The *t*-statistics are based on standard errors clustered at the firm level. The *t*-statistics are based on standard errors clustered at the firm level. ***, **, * denote statistical significance at the 0.01, 0.05, and 0.10 levels (two-tailed), respectively.

Panel A: Analyst Beats AI				
Variables				
<i>Amihud Illiquidity</i>	0.252*** (11.51)	0.218*** (6.06)	0.200*** (6.39)	0.119** (2.24)
<i>Intangible Assets</i>	0.027*** (4.09)	0.026*** (4.57)	0.021** (2.49)	0.023*** (2.88)
<i># Analysts in Brokerage Firm</i>	0.700*** (5.28)	0.931*** (7.14)	0.109 (0.29)	0.501 (1.34)
<i>Star Analyst</i>	-0.106 (-0.17)	-0.290 (-0.48)	0.373 (0.51)	0.207 (0.29)
<i># Information Events</i>	-0.026*** (-5.81)	-0.012*** (-2.68)	-0.025*** (-5.47)	-0.010** (-2.16)
<i>Distance to Default</i>	-0.009*** (-8.16)	-0.005*** (-3.75)	-0.014*** (-12.55)	-0.011*** (-8.41)
<i>Market Cap</i>	-0.071*** (-15.18)	-0.049*** (-10.84)	-0.085*** (-16.79)	-0.055*** (-10.79)
<i>% Institutional Holdings</i>	-0.033** (-1.99)	-0.011 (-0.58)	-0.012 (-0.70)	-0.004 (-0.43)
<i>Fluidity</i>	0.243* (1.89)	-0.273** (-2.02)	0.529*** (4.13)	0.032 (0.24)
<i>Text Complexity</i>	-0.001 (-0.67)	0.001 (0.65)	-0.003 (-1.25)	0.000 (0.09)
<i>Industry Recession</i>	0.012** (2.07)	0.030*** (5.06)	0.027*** (4.28)	0.044*** (6.84)
<i>Time Trend</i>	0.010*** (11.90)		0.009*** (8.60)	
Year Fixed Effect	No	Yes	No	Yes
Firm Fixed Effect	Yes	Yes	Yes	Yes
Analyst Fixed Effect	No	No	Yes	Yes
Observations	352,358	352,358	352,358	352,358
Adjusted R-squared	0.07	0.09	0.13	0.15
Panel B: Forecast Error Difference: Analyst vs. AI				
Variables				
<i>Amihud Illiquidity</i>	0.292*** (8.60)	0.263*** (11.03)	0.205*** (5.52)	0.129*** (3.71)
<i>Intangible Assets</i>	0.027*** (3.92)	0.026*** (4.04)	0.024*** (3.03)	0.025*** (3.31)
<i># Analysts in Brokerage Firm</i>	0.762*** (5.47)	0.990*** (7.14)	0.435 (1.07)	0.787* (1.95)
<i>Star Analyst</i>	0.105 (0.16)	-0.082 (-0.13)	0.436 (0.57)	0.262 (0.35)
<i># Information Events</i>	-0.022*** (-4.30)	-0.013** (-2.35)	-0.023*** (-4.37)	-0.011** (-2.04)
<i>Distance to Default</i>	-0.008*** (-6.41)	-0.004*** (-2.82)	-0.013*** (-10.09)	-0.010*** (-6.91)
<i>Market Cap</i>	-0.068*** (-13.76)	-0.047*** (-9.23)	-0.077*** (-14.62)	-0.049*** (-8.71)
<i>% Institutional Holdings</i>	0.031*** (2.77)	0.050* (1.83)	0.056*** (5.71)	0.061*** (3.81)
<i>Fluidity</i>	0.117 (0.76)	-0.383** (-2.32)	0.418*** (2.81)	-0.080 (-0.50)
<i>Text Complexity</i>	-0.000 (-0.16)	0.002 (0.67)	-0.001 (-0.45)	0.001 (0.54)
<i>Industry Recession</i>	0.014** (2.00)	0.030*** (4.33)	0.027*** (3.53)	0.042*** (5.56)
<i>Time Trend</i>	0.009*** (9.35)		0.008*** (7.02)	
Year Fixed Effect	No	Yes	No	Yes
Firm Fixed Effect	Yes	Yes	Yes	Yes
Analyst Fixed Effect	No	No	Yes	Yes
Observations	352,358	352,358	352,358	352,358
Adjusted R-squared	0.10	0.12	0.16	0.17

Figure 6: Table 4: determinants of analyst relative advantage

5.4 Table 5 and Table 6: disagreement and analyst incremental value

Read:

- Table 5 shows large analyst–AI disagreement is more likely for illiquid, intangible, smaller firms with less abundant information and higher distress risk.
- Conditional on high disagreement, humans are more likely to win for the same types of firms where institutional knowledge is valuable.
- Table 6 shows the hybrid model benefits more from analysts for illiquid, intangible, smaller, distressed,

and recession-exposed firms.

- Institutional ownership is positively related to the hybrid's relative performance, suggesting analyst information is more valuable where sophisticated investors are active.

Economic message: Disagreement is not noise. It identifies cases where the analyst may know something different from the machine. The hybrid gains most when human information is distinct from AI's public-data signal.

Table 5

Disagreement between man and machine.

This table presents the coefficients and t-stats of regressing the *Disagreement* indicator on the firm-level, industry-level, and macroeconomic variables presented in Table 1. For each pair of forecasts, we define the indicator variable *Disagreement* to be one if the magnitude of the difference between the absolute prediction errors of the analyst and our AI model, normalized by the maximum value of these two prediction errors, is above the 90th percentile among all forecasts on the same firm within three years. Conditional on *Disagreement* being positive, we further a sub-indicator, *Human wins*, which equals 1 if human analyst has a lower absolute prediction error. We report regression results with *Human wins* in the subsample where *Disagreement* being positive. To calculate t-statistics, standard errors are clustered at the firm level. ***, **, * denote statistical significance at the 0.01, 0.05, and 0.10 levels (two-tailed), respectively.

Dependent variable	Full sample		Disagreement sample	
	<i>Disagreement</i>		<i>Human wins</i>	
	(1)	(2)	(3)	(4)
<i>Amihud Illiquidity</i>	0.034*** (2.78)	0.034*** (2.74)	0.445*** (12.08)	0.383*** (8.02)
<i>Intangible Assets</i>	0.005*** (3.07)	0.005*** (2.87)	0.029** (2.05)	0.028** (2.18)
<i># Analysts in Brokerage Firm</i>	-0.605*** (-9.15)	-0.626*** (-9.39)	1.055*** (3.32)	1.209*** (3.86)
<i>Star Analyst</i>	-0.332 (-0.93)	-0.284 (-0.79)	4.129 (1.36)	5.603* (1.87)
<i># Information Events</i>	-0.007*** (-3.94)	-0.003 (-1.51)	-0.020** (-2.07)	-0.011 (-1.09)
<i>Distance to Default</i>	-0.005*** (-12.96)	-0.004*** (-8.55)	-0.002 (-0.94)	-0.002 (-0.90)
<i>Market Cap</i>	-0.003** (-2.94)	-0.005*** (-3.74)	-0.062*** (-8.26)	-0.041*** (-5.16)
<i>% Institutional Holdings</i>	-0.083*** (-5.50)	-0.080*** (-5.50)	-0.618 (-0.30)	2.335 (0.80)
<i>Fluidity</i>	0.138*** (3.56)	0.062 (1.59)	0.251 (1.10)	-0.217 (-0.93)
<i>Text Complexity</i>	0.000 (0.43)	0.001 (1.01)	-0.003 (-0.56)	-0.000 (-0.09)
<i>Industry Recession</i>	0.018*** (7.03)	0.018*** (6.75)	-0.006 (-0.50)	0.014 (1.09)
<i>Time Trend</i>	-0.000 (-0.18)		0.007*** (5.34)	
Year Fixed Effect	No	Yes	No	Yes
Firm Fixed Effect	Yes	Yes	Yes	Yes
Observations	352,358	352,358	42,023	42,023

(a) Table 5: disagreement between man and machine

Table 6

Man + Machine: The incremental value of analyst.

This table presents the coefficients and t-stats of regressing the *Analyst + AI Beats AI* indicator (Panel A) and *Forecast Error Difference: Analyst + AI vs. AI* (Panel B) on the firm-level, industry-level, and macroeconomic variables presented in Table 1. *Analyst + AI Beats AI* is an indicator variable equal to one if Analyst + AI beats AI. *Forecast Error Difference: Analyst + AI vs. AI* is defined as the difference between absolute prediction errors between AI and Analyst + AI, divided by the maximum value of these two prediction errors. The number is positive if the analyst has smaller absolute forecast error, i.e., Analyst + AI beats AI.

Panel A: Analyst + AI Beats AI				
Variables				
<i>Amihud Illiquidity</i>	0.274*** (6.30)	0.256*** (4.31)	0.181*** (4.37)	0.142** (2.53)
<i>Intangible Assets</i>	0.021*** (3.39)	0.018*** (3.15)	0.020*** (2.65)	0.018** (2.54)
<i># Analysts in Brokerage Firm</i>	0.246* (1.93)	0.321** (2.54)	0.471 (1.32)	0.417 (1.17)
<i>Star Analyst</i>	-0.001 (-0.00)	-0.105 (-0.17)	-0.113 (-0.15)	-0.216 (-0.30)
<i># Information Events</i>	-0.012*** (-2.80)	-0.004 (-1.02)	-0.013*** (-2.89)	-0.003 (-0.62)
<i>Distance to Default</i>	-0.010*** (-10.53)	-0.003** (-2.34)	-0.013*** (-11.76)	-0.006*** (-4.35)
<i>Market Cap</i>	-0.025*** (-6.40)	-0.017*** (-4.37)	-0.036*** (-7.77)	-0.023*** (-4.86)
<i>% Institutional Holdings</i>	0.041 (1.25)	0.061*** (3.77)	0.075*** (3.19)	0.089*** (8.06)
<i>Fluidity</i>	0.344*** (3.16)	-0.076 (-0.67)	0.618*** (5.27)	0.167 (1.32)
<i>Text Complexity</i>	0.002 (1.23)	0.005** (2.41)	0.001 (0.41)	0.004* (1.69)
<i>Industry Recession</i>	0.024*** (4.33)	0.030*** (5.20)	0.031*** (5.03)	0.036*** (5.71)
<i>Time Trend</i>	0.004*** (6.54)		0.064*** (4.01)	
Year Fixed Effect	No	Yes	No	Yes
Firm Fixed Effect	Yes	Yes	Yes	Yes
Analyst Fixed Effect	No	No	Yes	Yes
Observations	352,358	352,358	352,358	352,358
Adjusted R-squared	0.04	0.05	0.09	0.10

Panel B: Forecast Error Difference: Analyst + AI vs. AI

Variables				
<i>Amihud Illiquidity</i>	0.233*** (14.38)	0.224*** (14.67)	0.223*** (9.63)	0.207*** (9.29)
<i>Intangible Assets</i>	0.012*** (2.95)	0.010*** (2.67)	0.012*** (2.71)	0.011** (2.54)
<i># Analysts in Brokerage Firm</i>	0.168* (1.92)	0.229*** (2.61)	0.456* (1.80)	0.563** (2.20)
<i>Star Analyst</i>	-0.088 (-0.20)	-0.150 (-0.35)	-0.018 (-0.04)	-0.083 (-0.16)
<i># Information Events</i>	-0.001 (-0.41)	-0.002 (-0.60)	-0.002 (-0.92)	-0.001 (-0.38)
<i>Distance to Default</i>	-0.005*** (-7.19)	-0.002*** (-2.73)	-0.007*** (-7.74)	-0.004*** (-4.08)
<i>Market Cap</i>	-0.012*** (-4.52)	-0.007** (-2.50)	-0.016*** (-5.05)	-0.008** (-2.34)
<i>% Institutional Holdings</i>	-0.007 (-0.44)	-0.001 (-0.11)	0.002 (0.24)	0.005 (0.97)
<i>Fluidity</i>	0.042 (0.52)	-0.090 (-1.03)	0.147* (1.81)	0.023 (0.26)
<i>Text Complexity</i>	0.002 (1.24)	0.003* (1.67)	0.002 (1.01)	0.003 (1.55)
<i>Industry Recession</i>	0.010** (2.54)	0.013*** (3.01)	0.014*** (3.01)	0.016*** (3.36)
<i>Time Trend</i>	0.002*** (3.89)		0.002*** (2.66)	
Year Fixed Effect	No	Yes	No	Yes
Firm Fixed Effect	Yes	Yes	Yes	Yes
Analyst Fixed Effect	No	No	Yes	Yes
Observations	352,358	352,358	352,358	352,358
Adjusted R-squared	0.07	0.07	0.10	0.11

(b) Table 6: incremental value of analysts in Man + Machine

5.5 Table 7 and Table 8: synergies and extreme errors

Read:

- Table 7 shows Man-Machine synergies are correlated with both human-advantage and machine-advantage environments.
- Synergy is stronger for illiquid firms, smaller firms, firms closer to default, and industry recession periods.
- Table 8 shows that at the 90th-percentile threshold, analyst-only extreme errors occur in 9.34% of cases and AI-only extreme errors in 7.84% of cases.
- Man + Machine avoids 90.72% of analyst-only extreme errors and 43.56% of AI-only extreme errors.

- The hybrid creates a new extreme error in only 0.13% of cases where neither side had one.

Economic message: The strongest benefit of combining humans and AI is not merely a small increase in average accuracy. It is the reduction of severe errors.

Table 7
Incremental effect of M+M over both man and machine. This table reports the synergy of Man + Machine over both Man and Machine. The M+M Square Error Residual is defined as follows: We regress the squared forecast error of M+M on 2 variables: the machine-alone squared forecast error and man-alone squared forecast error. By flipping the sign of residual of the regression, we obtain M+M Square Error Residual. The M+M Forecast Error Residual is similarly defined using absolute forecast errors instead of squared forecast errors. To calculate *t*-statistics, standard errors are clustered at the firm level. ***, **, * denote statistical significance at the 0.01, 0.05, and 0.10 levels (two-tailed), respectively.

	M+M squared error residual				M+M forecast error residual			
<i>Amihud Illiquidity</i>	0.375*** (5.11)	0.371*** (5.22)	0.398*** (3.96)	0.398*** (4.24)	0.218*** (3.34)	0.215*** (3.70)	0.248*** (2.80)	0.250*** (3.29)
<i>Intangible Assets</i>	0.001* (1.71)	0.001 (0.96)	0.000 (0.02)	-0.000 (-0.36)	-0.000 (-0.16)	-0.001 (-0.88)	0.000 (0.23)	-0.000 (-0.23)
<i># Analysts in Brokerage Firm</i>	-0.060*** (-3.69)	-0.072*** (-4.48)	0.011 (0.24)	-0.004 (-0.08)	-0.061*** (-3.28)	-0.074*** (-4.08)	0.036 (0.74)	0.026 (0.57)
<i>Star Analyst</i>	-0.054 (-0.86)	-0.048 (-0.78)	0.005 (0.07)	0.009 (0.12)	-0.076 (-1.05)	-0.071 (-1.00)	-0.036 (-0.40)	-0.034 (-0.39)
<i># Information Events</i>	0.001** (2.25)	0.001** (2.57)	0.001** (2.07)	0.001** (2.41)	0.002** (3.14)	0.002** (2.54)	0.002** (2.80)	0.002** (2.46)
<i>Distance to Default</i>	-0.001*** (-7.77)	0.000 (0.26)	-0.001*** (-5.90)	0.000 (1.42)	-0.001*** (-8.22)	-0.001*** (-2.01)	-0.000 (-5.90)	-0.000 (-0.21)
<i>Market Cap</i>	0.006*** (7.81)	0.004*** (5.55)	0.005*** (6.48)	0.004*** (4.65)	0.005*** (7.83)	0.003*** (4.79)	0.005*** (6.78)	0.003*** (4.19)
<i>% Institutional Holdings</i>	-0.010*** (-2.75)	-0.010*** (-2.16)	-0.015*** (-6.82)	-0.014*** (-4.77)	-0.009** (-2.12)	-0.009** (-1.67)	-0.011*** (-5.31)	-0.011*** (-3.30)
<i>Fluidity</i>	0.010 (0.67)	0.026 (1.52)	0.008 (0.50)	0.020 (1.01)	0.007 (0.46)	0.040** (2.24)	-0.007 (-0.38)	0.025 (1.20)
<i>Text Complexity</i>	0.000 (0.03)	0.000 (0.37)	0.000 (0.06)	0.000 (0.44)	0.000 (0.53)	0.000 (0.90)	0.000 (0.84)	0.000 (1.12)
<i>Industry Recession</i>	0.004*** (4.68)	0.003*** (3.23)	0.004*** (3.95)	0.002*** (2.59)	0.004*** (4.43)	0.002*** (2.60)	0.003*** (3.16)	0.001 (1.54)
<i>Time Trend</i>	-0.001*** (-6.32)	-0.001*** (-3.85)	-0.000*** (-3.85)	-0.000*** (-3.85)	-0.001*** (-6.23)	-0.001*** (-4.47)	-0.001*** (-4.47)	-0.001*** (-4.47)
Year Fixed Effect	No	Yes	No	Yes	No	Yes	No	Yes
Firm Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Analyst Fixed Effect	No	No	Yes	Yes	No	No	Yes	Yes
Observations	346,301	346,301	346,301	346,301	346,301	346,301	346,301	346,301
Adjusted R-squared	0.13	0.14	0.23	0.24	0.08	0.09	0.15	0.16

Table 8
Probabilities of extreme errors: Man vs. Machine and Man + Machine. This table presents the probabilities of extreme errors by analysts and the AI and how the Man + Machine model helps to correct such errors. We benchmark the forecast error of each forecast to the 90th (or 75th, as a sensitivity check) percentile of squared prediction errors from all analysts on the same firm in the same year. Such a setup leads to four outcomes with regard to who committed an extreme error: (1) both the analyst and the AI model ("Both"); (2) Analyst; (3) AI; and (4) neither commits an extreme error ("Neither"). We examine these four scenarios and compute their empirical frequencies. We then compute the unconditional and conditional probabilities that the Man + Machine model can avoid the extreme error committed in the first three scenarios, and equally importantly, the probability that Man + Machine creates an extreme error in the fourth scenario. Panel A and B show results for extreme errors defined by the 90th percentile and 75th percentile of forecast errors, respectively.

Panel A: Probabilities of Extreme Errors (90th percentile)

	Both	Analyst	AI	Neither
Uncond. Prob.	3.46%	9.34%	7.84%	79.36%
Uncond. Prob.	0.16%	8.47%	3.41%	0.11%
Conditional Prob.	4.57%	90.72%	43.56%	0.13%

Panel B: Probabilities of Extreme Errors (75th percentile)

	Both	Analyst	AI	Neither
Uncond. Prob.	10.60%	16.82%	12.27%	60.32%
Uncond. Prob.	6.47%	14.39%	5.01%	0.10%
Conditional Prob.	4.39%	85.31%	40.86%	0.26%

(a) Table 7: incremental effect of Man + Machine

(b) Table 8: probabilities of extreme errors

5.6 Table 9: alternative-data event study

Read:

- Table 9 studies the introduction of alternative data, especially satellite-imaging data that create new consumer-footprint information.
- The key interaction $Treat \times AI\ Hiring$ is positive and statistically significant.
- The estimates are 0.533, 0.503, and 0.627 across the three specifications.
- The result means analysts improve relative to AI after alternative data become available only when their brokerage has AI capability.

Economic message: Alternative data alone are not enough. Humans catch up when their organizations can convert new data into usable forecasts through AI capability.

Table 9

Man + Machine event study: Alternative Data Coverage.

This table presents the coefficients and *t*-stats of regressing the *Analyst Beats AI* indicator on brokerage *AI Hiring*, *Alt Data Covered*, *Post*, and the interactions among these variables. *Analyst Beats AI* is an indicator variable equal to one if the analyst beats the AI. The AI and alternative data variables *AI Hiring*, *Alt Data Covered*, and *Post* are defined as follows. *AI Hiring* is the ratio of the number of AI jobs to the total number of job postings. *Alt Data Covered* is an indicator variable equal to one if alternative data are available for the firm by the end of the sample, and if the firm is in an industry with retail footprints. *Post* is an indicator variable equal to one if a “treated” firm has been covered by alternative data. For “untreated” firms, *Post* is coded one if the year is after 2014. The control variables are the firm-level, industry-level, and macroeconomic variables presented in Table 1. Standard errors are clustered at the firm level. ***, **, * denote statistical significance at the 0.01, 0.05, and 0.10 levels (two-tailed), respectively.

Variables	Analyst Beats AI		
<i>Treat</i> × <i>AI Hiring</i>	0.533** (2.16)	0.503** (2.04)	0.627*** (2.71)
<i>Treat</i> : <i>Alt Data Cover</i> × <i>Post</i>	0.002 (0.05)	0.028 (0.61)	0.029 (0.64)
<i>AI Hiring</i>	0.032 (0.90)	0.069 (1.52)	0.018 (0.52)
<i>Alt Data Cover</i>	0.029 (0.69)	−0.007 (−0.17)	
Year Fixed Effect	Yes	Yes	Yes
Firm Fixed Effect	No	No	Yes
Analyst Fixed Effect	No	Yes	No
Observations	56,697	56,697	56,697
Adjusted R-squared	0.03	0.07	0.14

Figure 9: Table 9: Man + Machine event study using alternative-data coverage

6 Short summary

- The paper builds an AI analyst for 12-month stock-return prediction and compares it with human analysts.
- The AI forecast uses public firm, industry, macro, textual, return, and earnings information, but excludes analyst forecasts.
- AI wins the average Man vs. Machine contest: humans beat AI only 45.5% of the time.
- Machine-debiased analyst forecasts improve to 46.5%, implying that correctable biases explain only part of the gap.
- AI-driven portfolio signals earn positive long-short returns and alphas relative to analyst signals.
- Humans do better where institutional knowledge matters: illiquid firms, intangible firms, small firms, distressed firms, and recessionary industry environments.
- AI does better when public information is abundant and voluminous.
- The Man + Machine hybrid outperforms both stand-alone analysts and stand-alone AI.
- The hybrid especially reduces extreme forecast errors.
- Alternative-data evidence suggests human analysts adapt when their brokerage houses develop AI capabilities.
- The paper’s central message is not replacement, but task-specific complementarity between human judgment and machine prediction.

7 Conclusion

7.1 Core conclusion

Cao, Jiang, Wang, and Yang show that an AI analyst can beat most human stock analysts in return prediction, but the victory is not complete. Humans still add value when firm information is ambiguous, contextual, or difficult to quantify. The best forecasting system is therefore not Man alone or Machine alone, but Man + Machine.

7.2 Economic intuition

AI excels at processing large amounts of structured and semi-structured public information. Human analysts excel when qualitative institutional knowledge, firm-specific context, and judgment under uncertainty are important. When the AI forecast and the analyst forecast are combined, the hybrid can use the strengths of both and avoid the worst errors of either side alone.

7.3 Takeaway

The paper provides a finance-based view of AI adaptation. Skilled workers are vulnerable when their tasks are mostly transparent prediction problems with many data inputs. But they remain valuable when their knowledge is complementary to AI. The future equilibrium is likely to reward analysts and organizations that learn to use AI as an augmentation technology.